

Next-Day Stock Status Prediction in Fresh Retail

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Problem Statement & Objective Metrics

Operational Challenge in Fresh Retail

- Perishable fresh items suffer from **rapid intra-day demand shocks** and supply replenishment lags.
- Unpredicted stockouts cause direct financial revenue loss, stock spoilage, and customer churn.

Predictive Target Formulation

- Forecast a **24-dim binary hourly status vector**:
 $\mathbf{Y} = [y_1, y_2, \dots, y_{24}]^T \in \{0, 1\}^{24}$
- Covers active operational store hours (**6 AM to 10 PM**).
- $y_h = 1$ indicates SKU stockout during hour h .

Key Objective Performance Metrics

- **Hour-Level F1-Score** (Primary Metric):
Evaluates precision vs. recall on positive stockout hours: $F1 = \frac{2 \cdot P \cdot R}{P + R}$
- **Stockout Recall**: Measures percentage of true operational stockout hours correctly detected ($R = \frac{TP}{TP + FN}$).

Taxonomy of Time-Series Forecasting Literature

- **Recurrent Sequence Memory** (Hochreiter & Schmidhuber, 1997): *LSTM* maintains persistent cell state memory C_t for temporal sequence continuity.
- **Deep Time-Series Transformers** (Lim et al., 2021 [TFT]; Zeng et al., 2023 [PatchTST]): Self-attention mechanisms suffer from **overparameterization** ($> 280k$ params) on short sequences.
- **Linear Series Decomposition** (Zeng et al., 2023 [DLinear]; Challu et al., 2023 [N-HiTS]): Moving-average trend decomposition provides parameter-efficient baselines ($< 15k$ params).
- **Explainable AI in Retail** (Sundararajan et al., 2017 [IG]; Lundberg & Lee, 2017 [SHAP]): Attribution methods reveal feature-level importance to guide neural architecture design.
- **FreshRetail Benchmark Studies** (Same Dataset): Recent domain-specific works like "*A Coupled Hierarchical Architecture for Multi-Granularity Demand Forecasting*" and "*When Less Data Is Better: Eliminating Stockout-Induced Bias in Choice Estimation*" use this dataset to model complex stockout biases.

FreshRetailNet-50K Dataset Overview

Dataset Scale & Scope

- **Volume:** 50,000 store-product pairs across 898 stores.
- **Timeframe:** 90-day series of highly granular hourly data.
- **Products:** 865 diverse perishable SKUs.

Rich Feature Set

- **Sales & Stock:** Hourly sale amounts & exact stock status.
- **Events:** Discount rates, holiday flags, and activities.
- **Weather:** Precipitation, temp, humidity, and wind.

Why We Selected This Dataset

- **Pioneering Benchmark:** It is the first large-scale public dataset specifically designed for **censored demand estimation** (where out-of-stock items hide true customer demand) in the fresh retail domain.
- **Perishable Goods Focus:** Perfectly captures the unique, rapid intraday volatility of short-shelf-life items, containing ~20% organically occurring stockout data.

Data Selection & Splitting Strategy

Unit of Analysis & Filtering

- **Target Grain:** Predicted at the level of `city_id + store_id + product_id`.
- **Filtering Logic:** Fresh retail datasets contain thousands of dormant items. We select the **Top 15 SKUs** to focus on high-impact, stockout-prone goods.

SKU Statistical Ranking Formula

$$\text{Score} = 0.40 \times \text{Sales} + 0.25 \times \text{Stockout Hours} + 0.20 \times \text{Stockout Days} + 0.15 \times \text{Sales Std}$$

Train/Validation Split

- **Chronological Split:** Strict temporal separation to prevent data leakage.
- **Training Set:** Historical data up to $T_{\max} - 15$ days (~ 2.5 months).
- **Validation Set:** Final 15 days of data.
- **Standardization:** `StandardScaler` fitted strictly on the training set to prevent lookahead bias.

Hourly vs. Daily Forecasting Paradigm

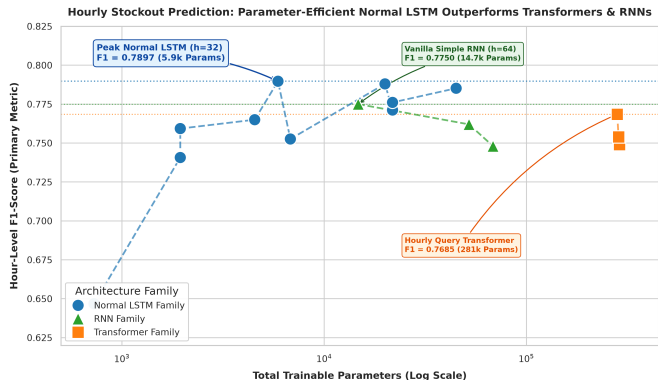
Traditional Daily Aggregation Paradigm

- **Input:** $L = 14$ days of historical daily metrics.
- **Output:** 1 scalar value (Total expected daily sales or binary stockout flag).
- **Loss of Information:** Store managers cannot determine *when* replenishment must occur.

Our 24-Hour Hourly Binary Paradigm

- **Input:** $L = 14$ days of SKU-store daily feature vectors.
- **Output:** 24 binary probabilities $\hat{\mathbf{Y}} \in [0, 1]^{24}$.
- **Actionable Insight:** Identifies peak risk hours (e.g. 11 AM lunch rush & 6 PM evening peak) for proactive restocking.

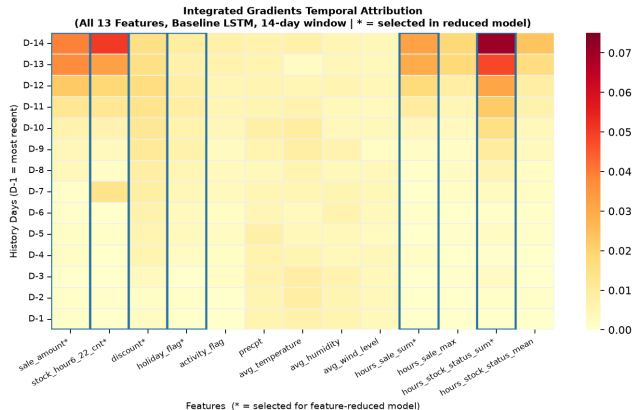
Model Family Comparison: Simple LSTMs Beat Heavy Transformers



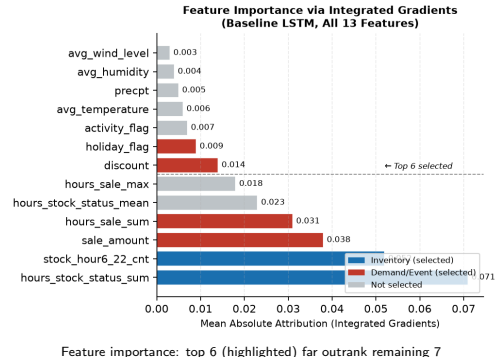
Key Empirical Findings

- **Parameter Efficiency:** Normal LSTMs (5.9k–44.9k params) reach **0.7852–0.7897 F1-Score**, matching or beating heavy Transformers.
- **Transformer Overparameterization:** PatchTST (285k params) & TFT (288k params) peak at **0.7540–0.7685 F1-Score**.
- **Why LSTMs Win:** Sequential state memory h_t naturally aligns with continuous hourly depletion better than permutation-invariant self-attention.

ML Explainability: Integrated Gradients Analysis



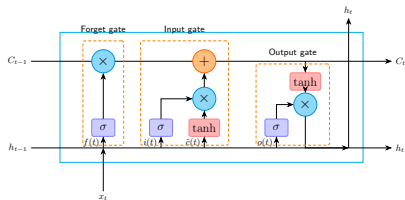
Attribution heatmap: all 13 features & 14-day history (D-1 = most recent)



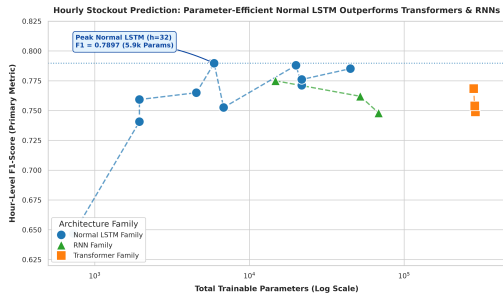
Observation → Selection

Out of all 13 features, IG reveals the same 6 features consistently dominate across time steps and samples. Weather & activity features show near-zero attribution. This motivates our **feature reduction to 6 signals** and the dual-stream split between demand & inventory.

Limitations of Baseline LSTM



Standard LSTM Cell Architecture

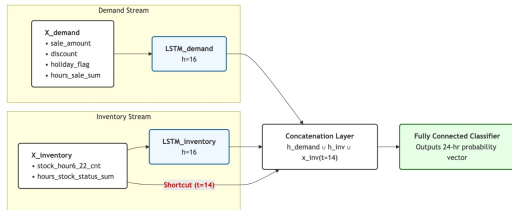


F1-Score across LSTM hidden size variants ($h=8$ to $h=96$)

Why Scaling Alone Doesn't Help

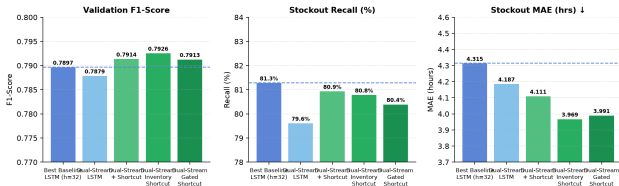
- **F1 plateau** at ~ 0.789 regardless of hidden size — more params yield no gain.
- Demand & inventory share hidden state $h_t \Rightarrow$ **feature cross-talk**.
- IG analysis (prev. slide) proves these signals need **dedicated streams**.

Our Fix: Dual-Stream Inventory Shortcut LSTM



Dual-Stream: Demand LSTM + Inventory LSTM + Day-14 Shortcut

LSTM Architecture Evolution: Baseline → Dual-Stream Variants



F1, Recall & MAE: all dual-stream variants vs. baseline LSTM

Key Results

- **Inventory Shortcut:** F1 0.7926, MAE 3.97 hrs at just 4,600 params.
- Decoupled streams consistently beat all baseline LSTM sizes.

Exploring Linear Decomposition: DLinear Models

Why Explore Linear Decomposition?

DLinear (Zeng et al., 2023) demonstrated that single-layer linear models decomposing time series into Trend and Seasonal components can outperform complex deep networks.

DLinear Series Decomposition

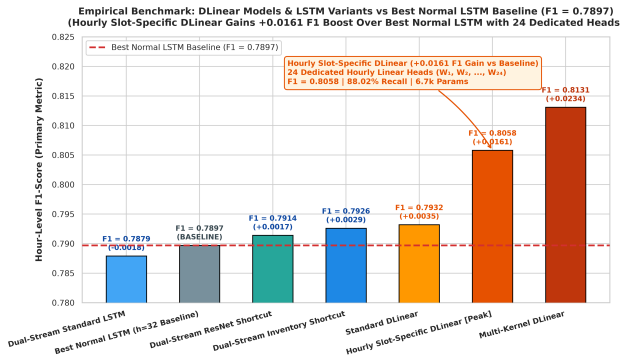
$$X_{\text{trend}} = \text{AvgPool}(X), \quad X_{\text{seasonal}} = X - X_{\text{trend}}$$

$$Y_{\text{pred}} = W_{\text{trend}}X_{\text{trend}} + W_{\text{seasonal}}X_{\text{seasonal}}$$

Standard DLinear Performance

- **Parameters:** 6,768 parameters.
- **Validation F1-Score: 0.7932**
 - Outperforms Normal LSTMs (**0.7897**)
 - Outperforms Enhanced LSTMs (**0.7926**)
 - Outperforms Transformers (**0.7540**)
- **Stockout Recall: 87.44%.**

Architectural Innovation 2: Linear Changes in DLinear



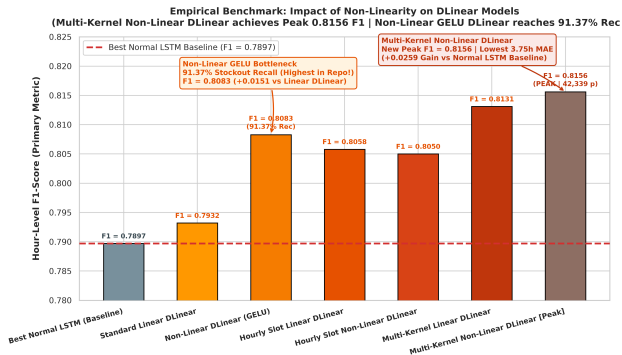
Hourly Slot-Specific Linear DLinear

- Uses **24 dedicated linear heads** ($W_1, \dots, W_{24}$) for each operational hour.
- Resolves inter-hour gradient conflict between morning and evening rush hours.
- Boosts F1 to **0.8058** (+0.0161 over LSTM).

Multi-Kernel Linear DLinear

- Multi-scale moving average trend pooling ($k \in \{3, 5, 7\}$).
- Reaches **0.8131 F1-Score** with pure linear decomposition (13.5k params).

Architectural Innovation 3: Introducing Non-Linearity into DLinear



Non-Linear Activation Bottleneck

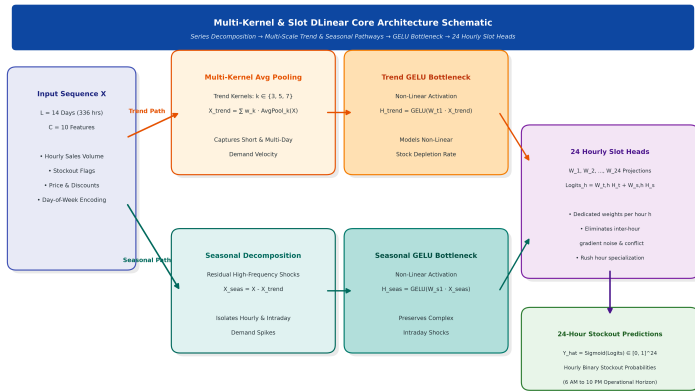
Adds GELU activation bottlenecks between decomposition and output projections:

$$\hat{Y} = W_{\text{trend},2} \text{GELU}(W_{\text{trend},1} X_{\text{trend}}) + \text{Seasonal}$$

Breakthrough Results

- **Non-Linear GELU DLinear:** Reaches **91.37% Stockout Recall** (Highest in Repo!) and **0.8083 F1**.
- **Multi-Kernel Non-Linear DLinear:** Reaches **0.8156 F1-Score** (New Peak F1 & lowest 3.75h MAE).

Architectural Schematic: Multi-Kernel & Slot DLinear



Core Mechanism

- **Trend Stream:** Multi-kernel moving average pooling ($k \in \{3, 5, 7\}$) + GELU bottleneck.
- **Seasonal Stream:** High-frequency residual stockout shocks ($X - X_{trend}$).
- **24 Slot Heads:** $W_1, \dots, W_{24}$ dedicated projections resolve inter-hour gradient conflicts.

Master Empirical Validation Leaderboard

Rank	Architecture Name	Category	Params	F1-Score	Recall	MAE (h)
1	Multi-Kernel Non-Linear DLinear	Non-Linear DLinear	42,339	0.8156	87.13%	3.75
2	Multi-Kernel Linear DLinear	Linear (Multi-Scale)	13,539	0.8131	85.22%	3.86
3	Non-Linear GELU DLinear	Non-Linear DLinear	21,168	0.8083	91.37%	4.20
4	Hourly Slot-Specific DLinear	Linear (24 Heads)	6,744	0.8058	88.02%	4.23
5	Standard Linear DLinear	Linear (Decomp)	6,768	0.7932	87.44%	4.28
6	Dual-Stream Inventory Shortcut	LSTM Family	4,600	0.7926	80.79%	3.96
Base	Best Normal LSTM (h=32)	Normal LSTM	5,912	0.7897	81.29%	4.31
7	Feature-Reduced Standard LSTM	Normal LSTM	19,992	0.7880	81.60%	4.58
8	Baseline Standard LSTM (h=96)	Normal LSTM	44,952	0.7852	80.40%	4.67
9	Vanilla Simple RNN (h=64)	RNN Family	14,744	0.7750	81.02%	4.23
10	PatchTST Patch Linear	Transformer	285,624	0.7540	81.90%	4.16

Conclusion & Operational Deployment Takeaways

Key Architectural Takeaways

- 1 **Non-Linearity Boosts DLinear:** GELU bottlenecks boost DLinear F1 to **0.8083** and Recall to a record **91.37%**.
- 2 **New All-Time Peak F1:** Multi-Kernel Non-Linear DLinear sets the highest F1-Score of **0.8156** and lowest MAE of **3.75 hours**.
- 3 **Feature Decoupling & Slot Heads:** Resolves inter-hour gradient noise and temporal cross-talk.

Deployment Recommendations

- **Primary Production Deployment:** **Multi-Kernel Non-Linear DLinear** (0.8156 F1, 3.75h MAE, 42.3k params).
- **Maximum Risk Mitigation:** **Non-Linear GELU DLinear** (91.37% Stockout Recall).
- **Future Scope:** Incorporate real-time intraday point-of-sale stream updates for dynamic threshold adjustment.